AGENTIC AI AND FRAMEWORKS

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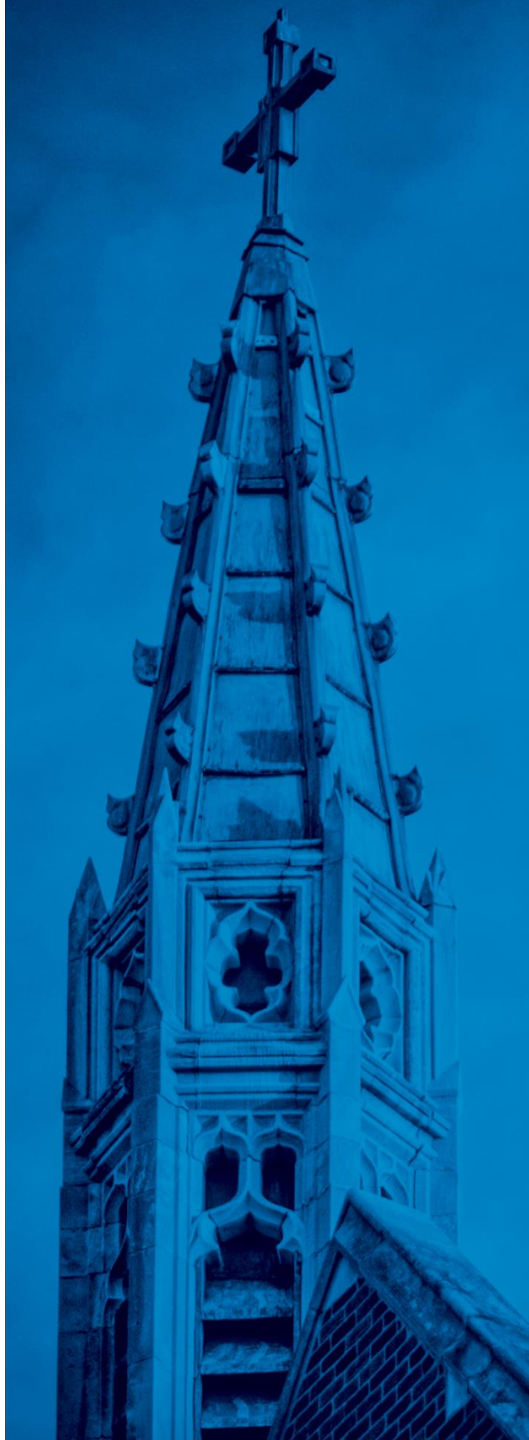
WHAT IS AGENTIC AI?

- **The AI "Brain" with Agency**
  - Unlike traditional AI that follows static instructions, **Agentic AI** is given a goal rather than a prompt.

  - It functions as an autonomous entity.



Reason

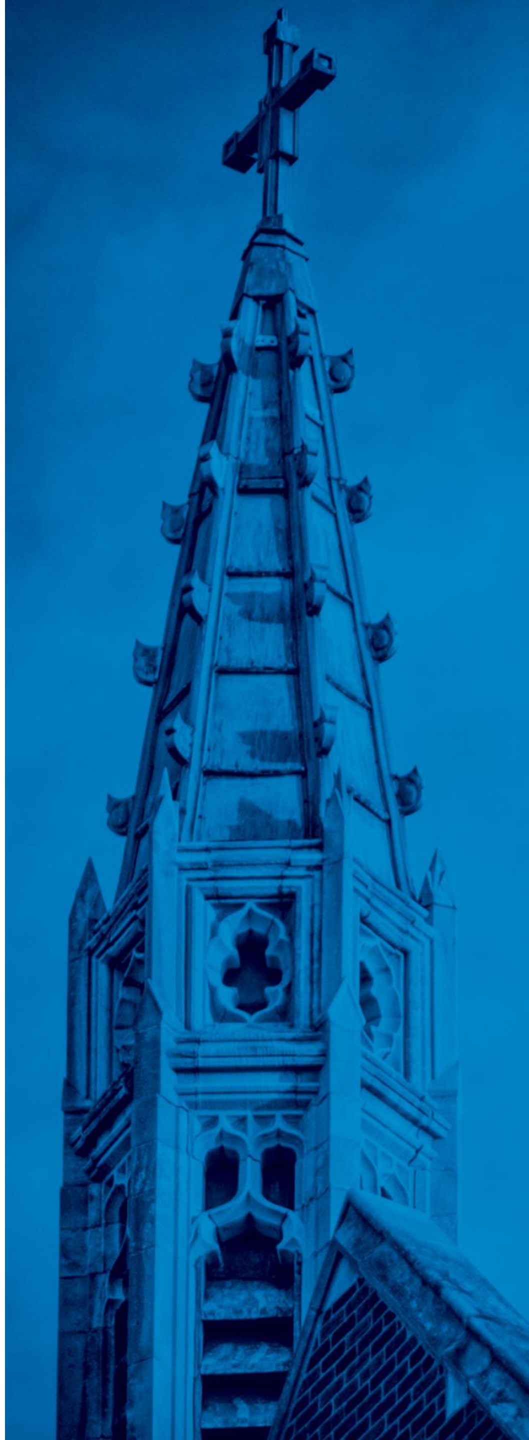
- Plan steps to reach a goal.

Use Tools

- Query APIs, Databases, or Web

Iterate

- Self-correct via feedback loops.

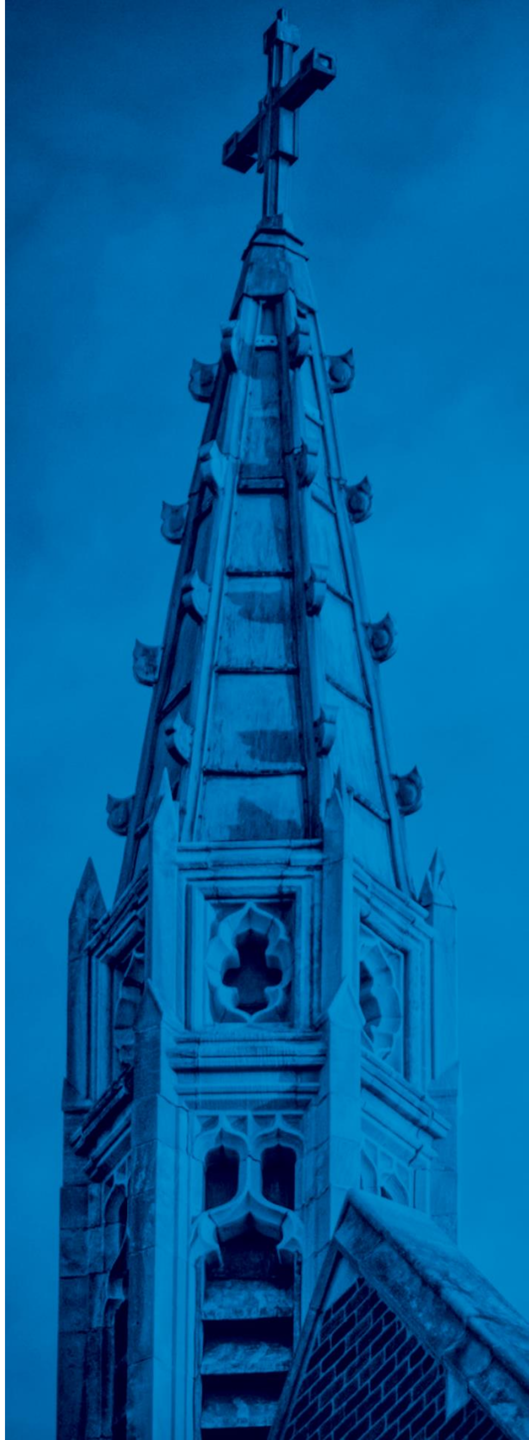


KEY PHILOSOPHY

- Work flow First, Goal over Instruction
  - Workflows as agent-orchestrated cycles

- A shift from telling AI "*how*" to do something (prompting) to telling it "*what*" to achieve.

- The agent manages the complexity of the execution path.



WHY? TO ADDRESS...

Hallucination

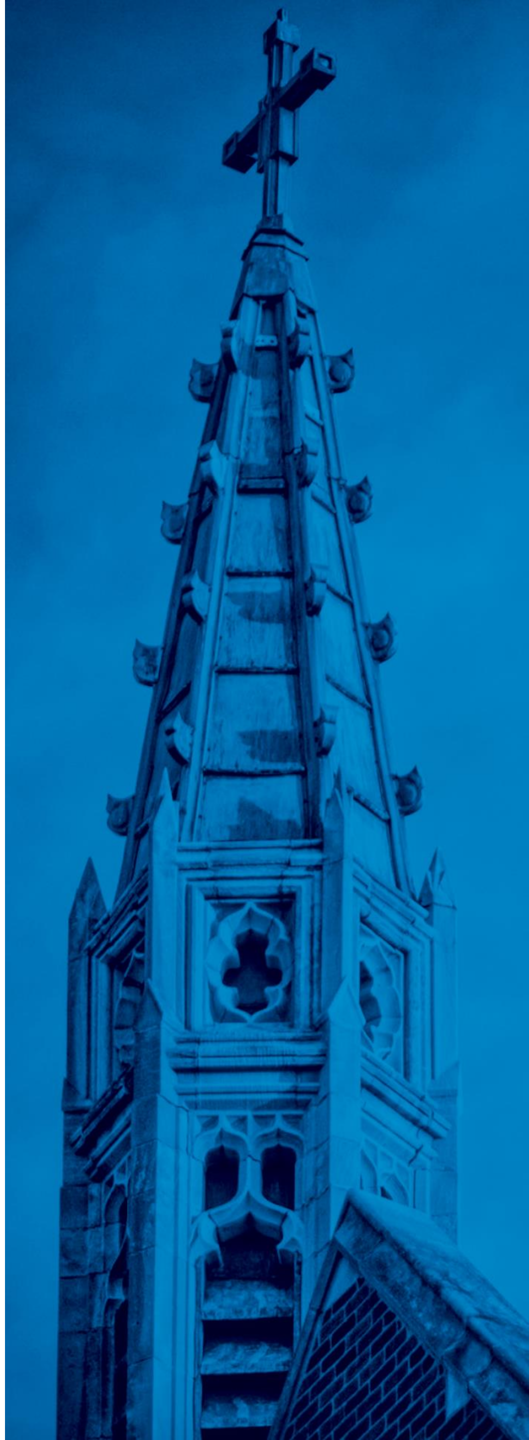
- Single-shot answers are prone to error.
- Agentic loops allow agents to "verify" their own work before presenting it to the user.

Isolation

- Traditional AI is trapped in a chat box.
- Agentic AI connects to files, internet, and enterprise APIs to effect real change.

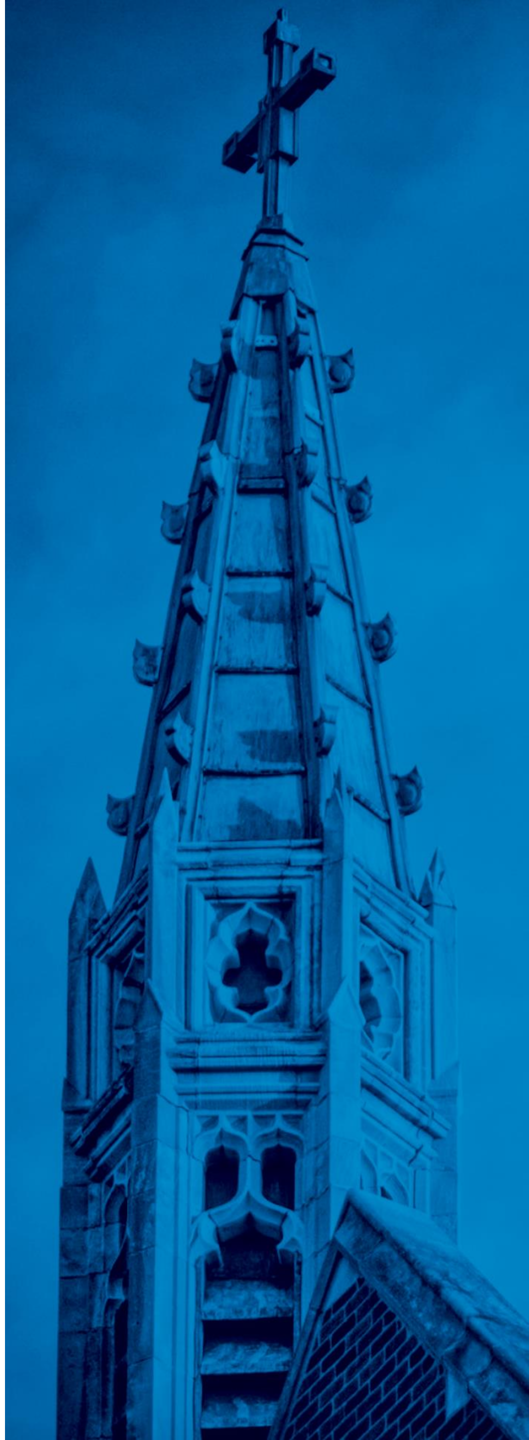
Complexity

- Multi-step tasks (e.g., tax filing or software debugging) are too large for one prompt.
- Agency breaks them into manageable loops.



A COMPARISION

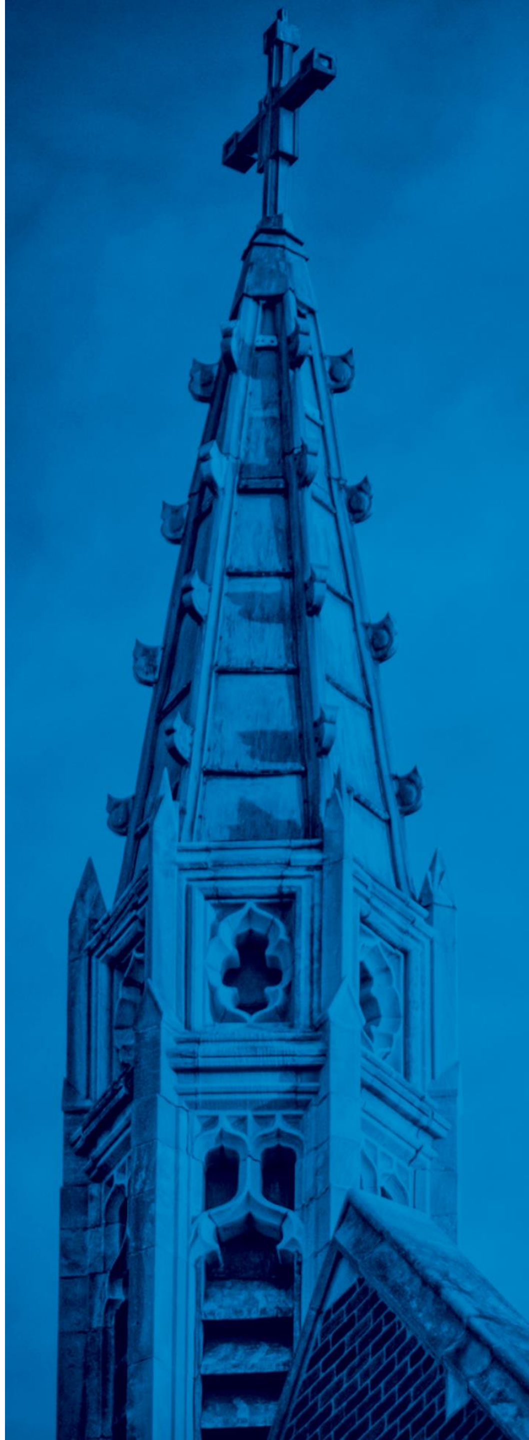
Feature	Traditional AI (Chatbots)	Agentic AI (Autonomous)
Interaction	Input → Output (One-shot)	Goal → Plan → Act → Result (Loop)
Autonomy	Follows specific prompts	Makes decisions on tool selection
Environment	Isolated within the UI	Connected (APIs, Web, DBs)
Error Handling	Hallucinates or fails	Self-corrects via internal feedback



MULTI-AGENTIC SYSTEM

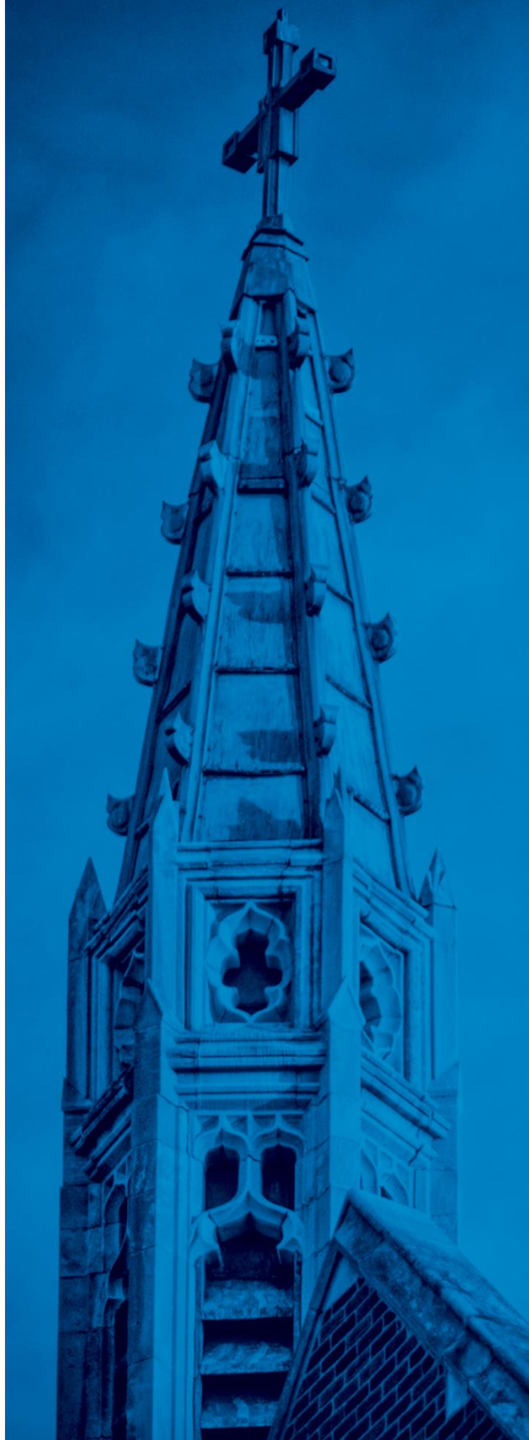
- Multi-Agent System - MAS is a framework where several specialized, autonomous AI agents collaborate, negotiate, or compete to solve complex, multi-step tasks.

- Unlike a single agent model, MAS assigns specific roles (such as researcher, writer, and editor) to different agents that communicate to achieve a shared goal



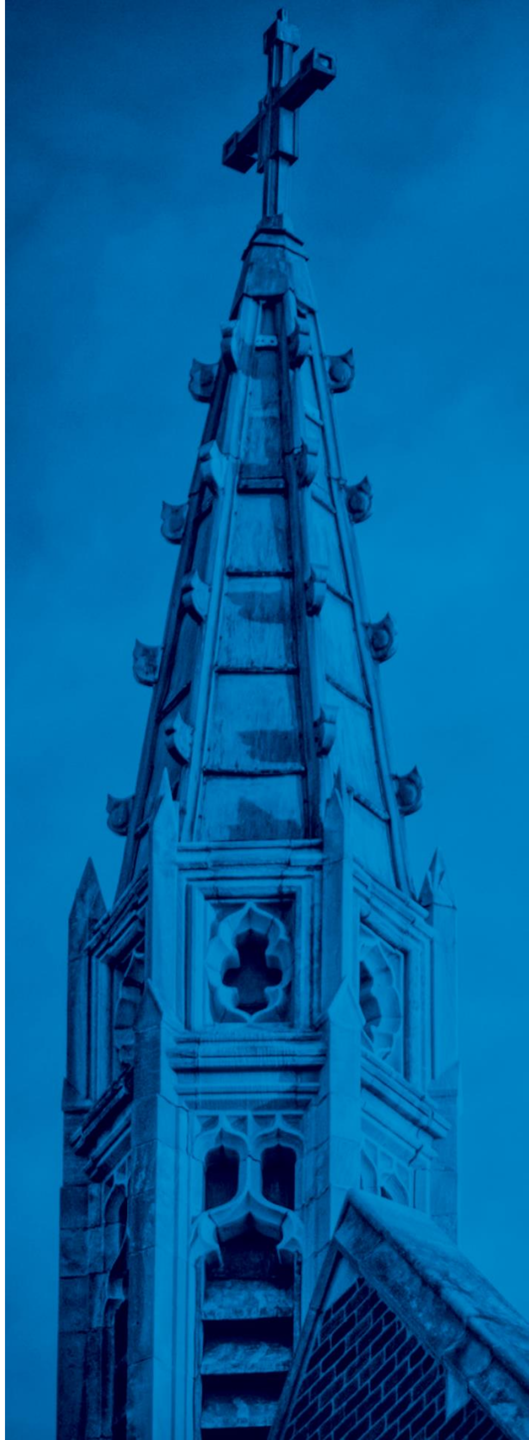
TYPICAL COMMUNICATION PATTERNS

Pattern	Logic	Communication Flow
Sequential	Agent A finishes its work and "hands off" the result to Agent B.	**Linear:** A → B → C
Hierarchical	A "Manager Agent" assigns tasks to "Worker Agents" and synthesizes their results.	**Top-Down:** Manager and Workers
Collaborative	Agents "see" a shared workspace and pick up tasks based on their specific skills.	**Mesh:** All agents talk to a shared "Blackboard."



AGENTIC AI FRAMEWORKS

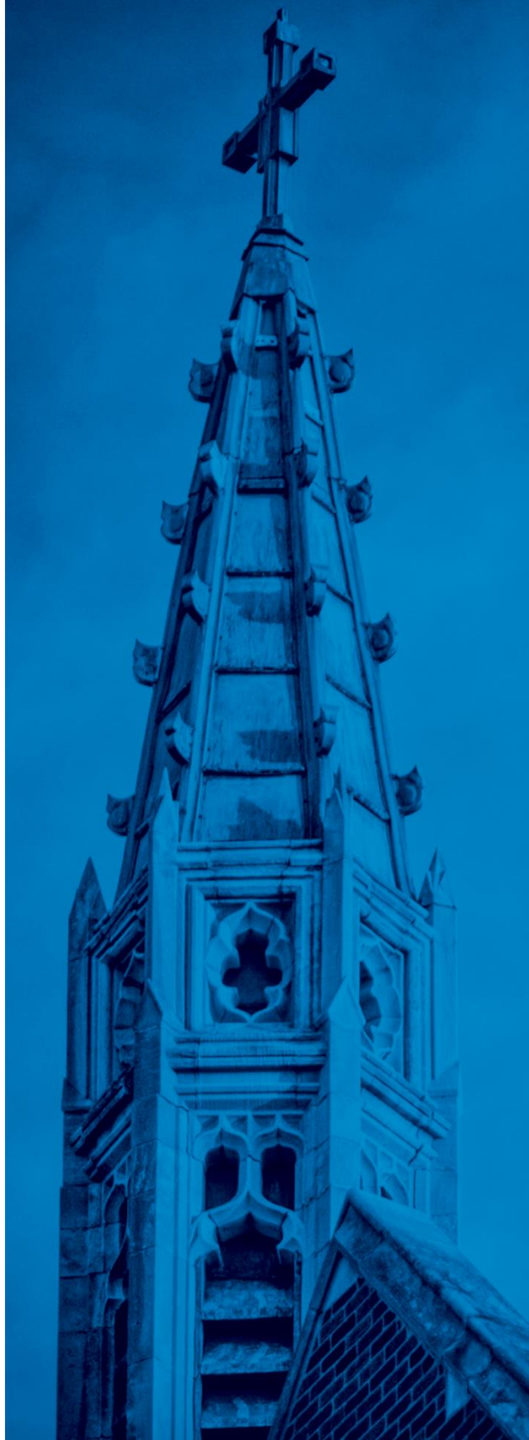
- **LangGraph** excels in **precise control and reliability** by modeling agent interactions as stateful cyclic graphs, making it the top choice for complex, custom workflows that require strict logic and state management.



AGENTIC AI FRAMEWORKS

- CrewAI stands out for its **intuitive, role-based orchestration**, allowing developers to quickly assemble "crews" of agents that mimic human organizational structures with minimal boilerplate code.

```
coder = Agent(
    role='Senior Software Engineer',
    goal='Write clean, efficient Python functions based on requirements.',
    backstory=(
        "You are a coding prodigy. Your focus is on writing modular, readable, "
        "and performant Python code. You take feedback seriously and can "
        "debug complex logic quickly."
    ),
    allow_delegation=False,
    verbose=True
)
```

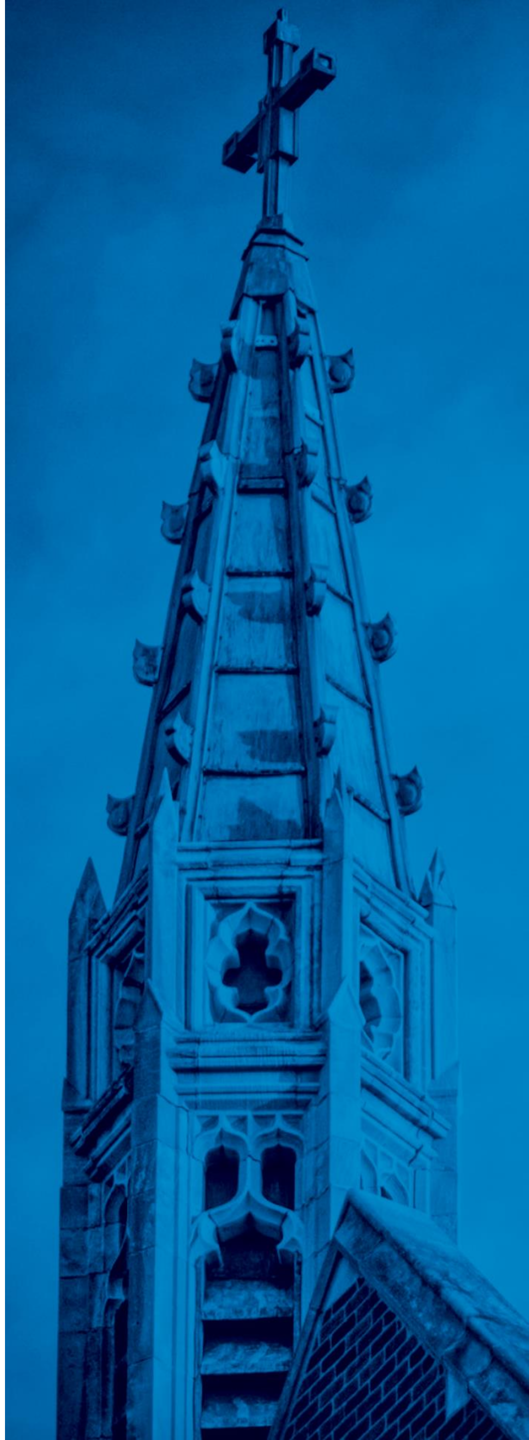


AGENTIC AI FRAMEWORKS

- Microsoft AutoGen is a leader in **conversational flexibility**, providing a robust framework for building multi-agent systems where agents can autonomously engage in dynamic, open-ended dialogues to solve problems.

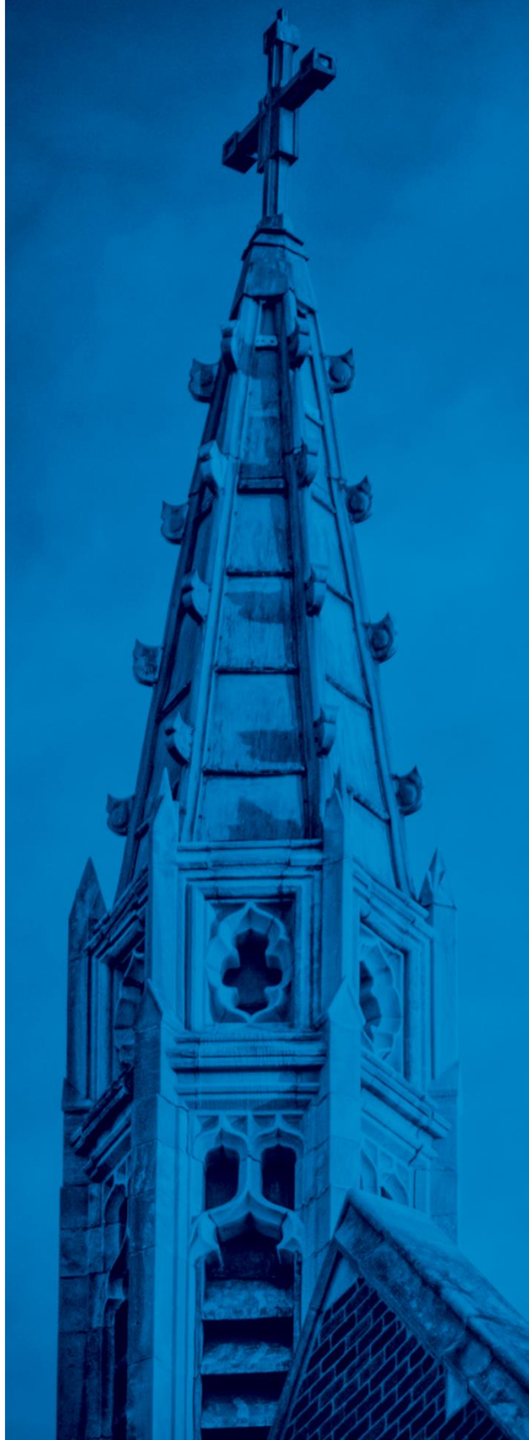
microsoft/**autogen**

A programming framework for agentic AI



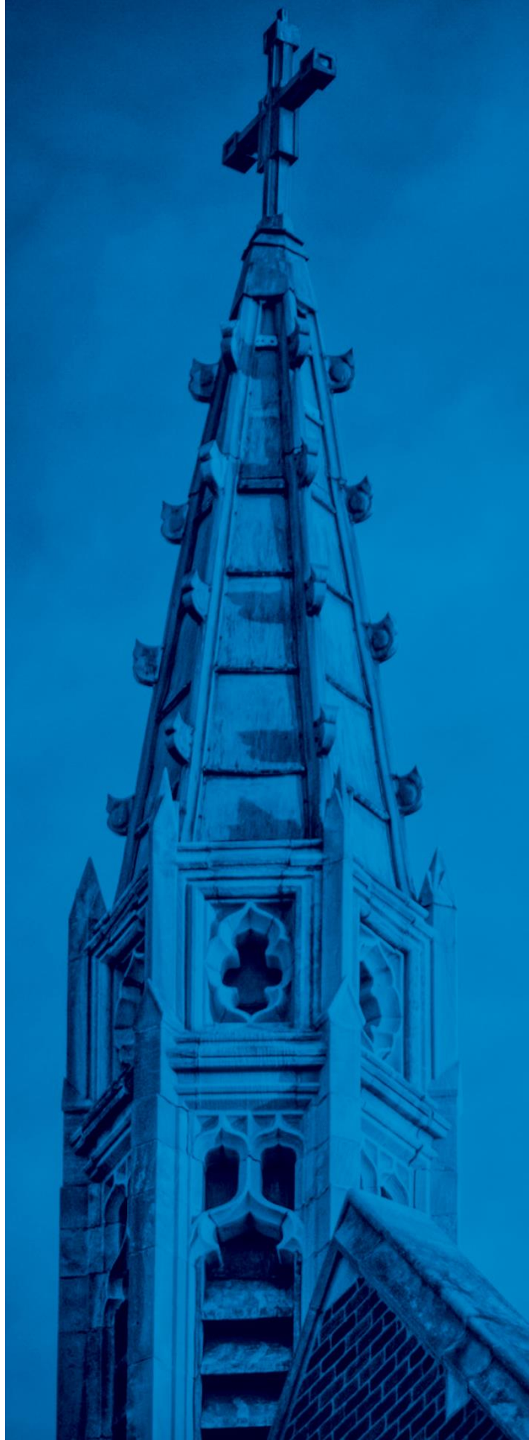
AGENTIC AI FRAMEWORKS

- **Google Vertex AI Agent Builder** offers **enterprise-grade scalability and integration**, leveraging Google Cloud's infrastructure to provide secure, distributed agent management and native interoperability.



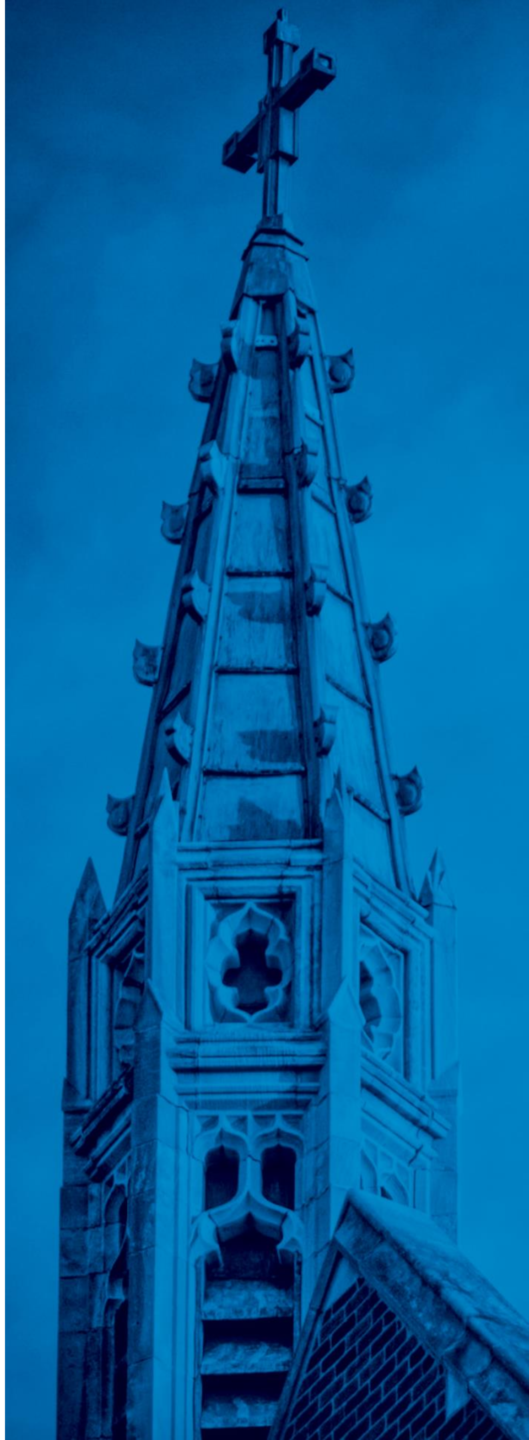
MULTI-AGENT SYSTEMS: APPLICATIONS

- **A. Cybersecurity: "The SOC (Security Operations Center) Crew"**
  - **Agent 1 (The Sentry):** Monitors network logs for suspicious IP addresses.
  - **Agent 2 (The Investigator):** When alerted by the Sentry, this agent cross-references the IP with global threat databases.
  - **Agent 3 (The Auditor):** Reviews the findings of the first two and decides whether to automatically update the firewall.
  - **Communication:** *Sequential.* If the Investigator finds a threat, it triggers the Auditor.



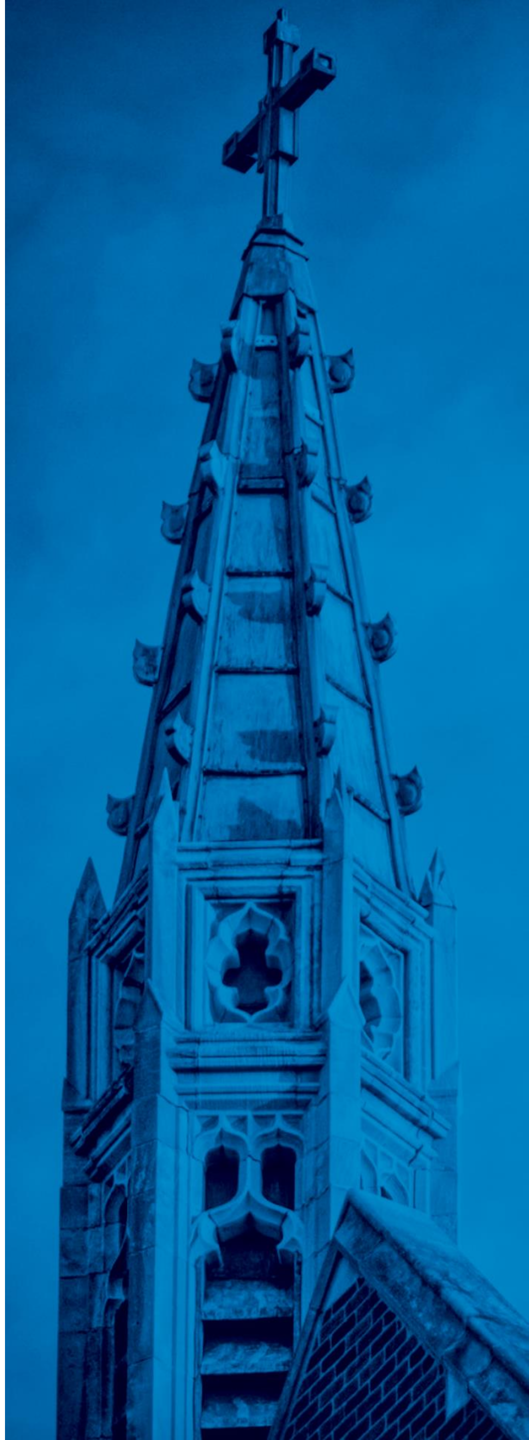
MULTI-AGENT SYSTEMS: APPLICATIONS

- B. Finance: "The Investment Committee"
  - **Agent 1 (The Sentiment Analyst):** Scans news and social media for a specific company.
  - **Agent 2 (The Quantitative Analyst):** Analyzes the company's 10-K financial filings and balance sheets.
  - **Agent 3 (The Strategist):** Compares the "vibe" (Sentiment) with the "numbers" (Quant) to suggest a Buy/Sell/Hold rating.
  - **Communication:** *Hierarchical.* The Strategist acts as the manager, requesting data from the other two.



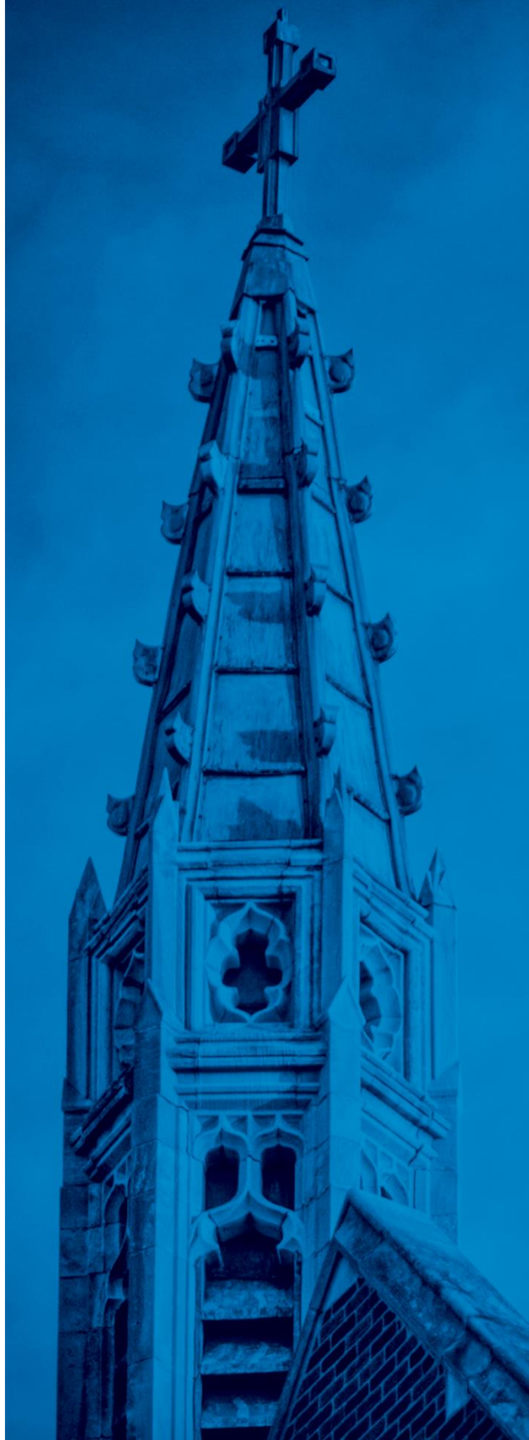
MULTI-AGENT SYSTEMS: APPLICATIONS

- C. Software Engineering: "The Autonomous DevOps Team"
  - **Agent 1 (The Coder):** Writes a Python function based on a user requirement.
  - **Agent 2 (The Tester):** Writes a unit test for that code and runs it.
  - **Agent 3 (The Critic):** If the test fails, the Critic sends the error log back to the Coder with specific instructions on how to fix it.
  - **Communication:** *Collaborative/Iterative*. They talk back and forth until the code is perfect.



MULTI-AGENT SYSTEMS: APPLICATIONS

- D. Data Analytics: "The Insight Generator"
  - **Agent 1 (The Data Cleaner):** Identifies missing values or outliers in a CSV file.
  - **Agent 2 (The Visualization Expert):** Suggests the best graph types to represent the clean data.
  - **Agent 3 (The Storyteller):** Takes the final charts and writes a 5-bullet point executive summary.
  - **Communication:** *Sequential.* Each step adds value to the data until it becomes an insight.



COMMUNITY INTEGRITY

- Communication Integrity.
  - **The Question:** If Agent A hallucinates and tells Agent B a lie, how does Agent B know?
  - **The Solution:** "Cross-Agent Verification." In a Multi-Agent system, we often set up an "Adversarial Agent" whose only job is to try to find errors in the other agents' communications. This is the AI version of **Peer Review**.